SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I K.N. Ritheesh (9925720971) (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Cloud Computing:

It is a technology that Provide Computing Resources such as Serves, Storage database, networking and solftware over the internet. Instead of Purchasing & Maintaining Physical hardware, users can Access these Resource from cloud Service Provide's whenever Required.

Ex: Amazon Web service, Microsoft Azure.

Four characteristics of cloud computing:

1. **On-Demand Self-Service:**

users can Access computing Resource's such as Storage, Virtual Machine and database whenever Required without Contacting the Service Provide.

2. **Resource Pooling:**

cloud Provides combine computing Resource to Serve Multiple customers.

Resource are dynamically, allocated According to user Requirements.

3. Broad Network Access:

Cloud Services are Accessible over the Internet Using different device Including Laptop's, Smartphones, tablet's and desktop Computer.

4. Rapid elasticity:

cloud Resources Can be Increased or decreased quickly acording to work load Requirements. this allow organization to handle Sudden increased or decreased in demand.

5. Measured Service:

Cloud Provider Monitor Resource usage, such as storage, Processing Power and bandwidth. Customer are generally charged.

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2) Evolution of cloud Computing:

Introduction:

It evolved through Several technological developments, beginning with

centralized computing and Processing through virtualization, Internet technologies & the distributed Infrastructure. These development made it Possible to deliver computing Resource's.

Evolution of cloud Computing:

1. Mainframe Computing (1950-1960s)

Mainframe Computer's Provided centralized Processing for May users through terminal's. Computing the Resource's were expensive so, Multiple user's shared the same Powerful System.

2. Personal and distributed system:

Minicomputer and Personal Computer's Made computing More Accessible organization's began distributing work Load Across Multiple Computer rather than replying entirely on one System.

3. Virtualization (1990's)

It enabled Multiple virtual Machine to Run on a single Physical

server. this improved hardware utilization and Provide the Foundation for flexible, isolated computing environments.

3) Server Virtualization in cloud Computing:

Serve Virtualization is a key technology Used in cloud Computing It allows one physical Serve to Run Multiple Virtual Machines, each with its own OS and allocation Resources. cloud Computing user's technologies such as Virtualization, Automation.

Role of Server Virtualization:

1. Efficient Resource virtualization:

Virtualization allows Served Virtual Machine to use the Processing Power, Memory and Storage of one Physical Serve.

2. Cost Reduction:

organization can Run Multiple workload's on fewer Physical Server's.

3. Isolation:

Each virtual Machine operate's in an Isolated environment. Problem's in one Virtual Machine can after be contained without directly affecting other virtual Machine.

4. Scalability & flexibility:

Virtual Machine can be Created, Modified, cloned or Removed Acording to application Requirements.

5. Rapid Provisioning:

New virtual server's can be created through software in Minutes Rather than Requiring the Installation.

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4) Role of Data in delivering cloud Service:

A data centre is a physically facility that Contains Server's storage System's. Networking equipment, Power

System's and Cooling Infrasture. Cloud Service Provide's Use data center's to store Data, Run Application's. Process Information..

Role of Data Centre's in Cloud Computing:

1. **Computing Power:**

Data Center's Contain Powerfu Servers that Process user Requests, execut Application and Perfor complex Calculation

2. **Data Storage and Management:**

Data Center's contain storage device that store database, Documents, images, videos, backup's and Application files.

3. **Network Connectivity:**

Routers, Switches and Communication links connect servers to the Internet and to other data center's.

4. **High Availability & Reliability:**

Cloud Provide use

Redundant hardware, backup system & Multiple facilities to Reduce Service interruptions.

5. Security:

Data centre use Physical Security, Access Controls, Surveillance, Firewalls encryption & Monitoring System.

Differentiate Iaas, Paas, Saas and Xaas.

Cloud computing service are commonly categorized According to the level of infrasture as a service, platom as a service, Software as a Service and Anything as a service.

1. Iaas:

It Provides fundoumental computing Resources such as virtual machine, storage and Networking over the Internet. Customers Manage their operating system

2. Paas:

It Provides a Platform for developing, testing, deploying and managing

Application is the Provider Manage Much of the Underlying Infrastructure, operating System and Runtime error.

3. **Saas**:

Saas delivers ready to use software over the internet. User's generally Access the application through a browser or app without installing or maintaining the Underlying Infrastructure.

4. **Xaas**:

Xaas is a broad term describing the delivery of Many kind's of technology capabilities as service over a network. It can includes Iaas, Paas and Saas as well as service such as Database as a DBaas.

Ex: MongoDB Atlas.

10/10/25